

(VII) Recourses Required

- Computer System with basic configuration. Linux/Ubuntu/ CentOS /any other open source Operating System.
- Software : Turbo C/ vi editor/ Any open source Python IDE such IDLE, Jupyter Notebook, Spyder, PyCharm etc.)

(VIII) Precautions to be followed

- (1) Handle computer system with care.
- (2) Strictly follow the instructions for writing, compiling, and executing the program.
- (3) Don't forget to save file before execution.
- (4) Follow safety Practices

(IX) Program Code :

Write a program on sequential file allocation method..

Note : Attached the code at the end.

This practical can be performed in any of the compiler like Turbo C/ vi editor etc or Python Interpreter / IDE like any open source IDE such IDLE, Jupyter Notebook, Spyder, PyCharm etc.)

(X) Conclusion

Successfully implemented C/Python program on Sequential File Allocation method

(XI) Practical Related Questions

Note: Below given are few sample questions for reference. Teachers must design more such questions to ensure the achievement of identified CO.

- (1) What are the various file allocation strategies?
- (2) What is Sequential file Allocation Method?
- (3) What are the disadvantages of sequential allocation methods?
- (4) Compare sequential allocation with linked and indexed file allocation methods.

[Space to Write for Answers]

1) The three main file allocation strategies are :-

1. Contiguous Allocation

Stores files in consecutive disk blocks.

2. Linked Allocation

Stores files blocks at different locations connected by pointers.

3. Indexed Allocation

Uses an index block to store the addresses of all file blocks.

2) Sequential (Contiguous) File Allocation is a method in which a file is stored in consecutive disk blocks. The operating system records the starting block and the file length, providing fast access but causing external fragmentation and making file expansion difficult.

3) 1. External Fragmentation

Free disk space becomes scattered into small blocks, making it difficult to allocate large contiguous space for new files.

2. Difficult File Expansion

If a file grows in size, there may not be enough contiguous free space available next to it.

3. Space Wastage

Extra space may be reserved for future file growth, leading to unused disk space.

4. File Relocation

If contiguous space is unavailable, the entire file may need to be moved to another location, which is time consuming.

4) Contiguous Allocation	Linked Allocation	Indexed Allocation
Stores files in consecutive disk blocks.	Stores file blocks anywhere on the disk and links them using pointers.	Uses a separate index block that stores the addresses of all file blocks.
Supports fast sequential and direct access.	Good for sequential access but slow for direct/random access.	Supports both sequential and direct/random access efficiently.
Causes external fragmentation	No external fragmentation	No external fragmentation
File expansion is difficult.	File expansion is easy.	File expansion is easy by updating the index block.
Simple to implement	Moderate complexity	More complex because of index management.
No pointer overhead	Requires pointer space in each block	Requires extra space for the index block.